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ORIGINAL PAPER

BIM-Enabled Structural Design: Impacts and Future Developments in Structural Modelling, Analysis and Optimisation Processes

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Abstract This review focuses on identifying the impacts and future development trends for current structural design practice in integration of building information modelling (BIM) technologies. BIM technologies, as novel information management schemes they are, are changing the conventional structure design processes. Currently, utilising BIM technologies for reshaping structural design has been widely acknowledged and embraced by both academic and industry circles. In this research, the current status and issues of the structural design processes (including modelling, analysis, and optimisation of structures) are fully investigated with emphases on specific design stages. The research put efforts on surveying the benefits of BIM in facilitating current structural design, such as systematic modelling processes, powerful interactive visualization platform, and standardised exchanging data interfaces. Impacts of personnel involvement in structural design when adopting BIM have also been identified in detail. Finally, a predicted cross-functional flowchart of BIM-enabled structural design for the near future is proposed, which shows future developing trends in improving structural design quality and addressing current issues.

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1 Introduction

As core concerns in structural design safety and cost issues, structural modelling, analysis and optimisation are focused on providing effective and economic solutions for sequential material fabrication and actual construction. These three processes represent essential state-of-the-art design steps. A general procedure of structural design starts up with conceptual design in line with design goal requirements, including owners' demands, structural codes, aesthetics and stability concerns. The drafted design then moves to the structural detailing phase with iterative evaluation processes and outputs a structural solution for implementation in construction. The structural design in this research represents all necessary procedures among current construction design practice, including conceptual and detail design processes of structures at architectural, structural and system integration perspectives. It contains surface and geometrical modelling, structural analysis and optimisation, sustainability analysis, etc. Among those processes, modelling, analysis and optimisation play significant roles in determining how well the final design fulfilled different deign intentions. The evaluation of structural design will also be continued at construction phase by raising change order requests.

Considering the development of structural modelling technologies, computer-aided modelling began to be developed over four decades ago in order to provide spatial information, to assist communication, and to increase the modelling efficiency of architecture projects based on digital models [1]. In the past ten years, the development of computer-aided tools has focused on assisting designers to finish their work effi-

ciently, increase flexible modification capacities, and eliminate design errors. As for structural analysis and optimisation, over five decades' efforts with the invention of finite element method (FEM) makes that analysis and optimisation schemes can be adopted naturally into the structural engineering practice [2]. The ever increasing demands on structure design efficiency and reliability have made structural optimisation a popular research area.

Current issues on the development of structural design usually involve the lack of flexibility, tedious or time-consuming labour works, and communication gaps with different design aspects. The modelling processes at conceptual, or detailed phases may fail to respond to various considerations from architectural and structural aspects, hence limiting the spaces for discovering superior design solutions at early stages. They also lack unified data sources that can be easily convert between architectural and structural models. It may result in inflexible design plans that may require significant reworks when redesign, or change order requests are raised. As for structural analysis and optimisation, they are non-interactive processes, and the result from analysis requires engineers' interpretation to adjust design directions, and as a result still requires skilled and experienced engineers to make decision on such circumstances to complete optimisation of processes to get better solutions. A significant amount of research focuses on solving these issues and aims to provide uniform solutions for them. With such trends of development, Building Information Modelling (BIM) technologies play an essential role for structural design practices and already show considerable benefits on solving those integration issues.

BIM technology, as a revolutionary approach for Architecture, Engineering, and Construction (AEC) domain since early 21st century, has been widely used in many countries in current construction practice. According to definition proposed by Eastman et al. [3], BIM represents a modelling technology and a set of processes to produce, communicate and analyse building models, including sustainable practice, management issues around technology, and assisting methods [4]. A building information model is expected to be an accurate virtual model that contains precise geometry, regulations, codes and relevant data to support the design, procurement, fabrication and construction activities for realising the construction results [3]. BIM technology is suitable to remove obstacles in designing structures, and is an appropriate media to store essential information for improving accuracy and efficiency of the structural design process. It can be seen in state-of-the-art cases demonstrated by recently research.

The general applications of current BIM technologies can be used for the following purposes: cost estimating, code reviews, visualization, fabrication/shop drawings, construction sequencing, conflicts and collision detection, facilities managements and so forth [5]. The advantages of implementing BIM have been studied and proved in many fields, such as reducing design and drafting errors, increasing productivity, flexible design, reducing design and change order costs, increasing communication between different construction teams, etc. With the rising amount of adoptions for BIM technologies, BIM-enabled structural design becomes even more popular, and is expected to boost more benefits, with cutting edge visualization technologies support in the next ten years.

BIM-enabled structural design is not merely an integrative process to combine two ideas for improving conventional construction, but represents a novel methodology, or a management guideline to design, model, and execute construction projects in practice, and by its nature it already contains the structural design elements at global aspects. The meaning of BIM-enabled structural design lies in a structural design process which embodies BIM spirits, such as obeying BIM modelling rules during architectural modelling processes, considering more factors other than geometric properties and so on. Literature reviews suggests that BIM is most suitable to be used in design and pre-construction stages [6], and it has potential to be improved in the expecting future.

The purpose of this research is to identify impacts and potential developments for the structural design domain with the consideration of BIM, by surveying current practices of such two topic integration. In particular, the perspectives for identifying the impacts and future development trends in this research are based on personnel involving in design processes. The remainder of this paper is organised as follows: in Sect. 2, the state-of-the-art structural design procedures are introduced, including general work flow of practice in construction projects; three parts of processing steps: parameterised modelling, structural analysis and optimisation, and sustainability evaluation, are emphasised in detail, including current issues of development in order to show current status and motivations of implementing BIM. Section 3 then introduces BIM adoptions in designing structural. Section 4 identifies the impacts of embracing BIM throughout designing, modelling and optimising structural processes. Future developments of BIM-enabled structural design are proposed in Sect. 5, which provides research opportunities for improving conventional design processes and suggesting the directions involved. Conclusions are given in the last section.

2 Structural Design Practices

Structural design is defined as sequence design actions related to structures, including their appearance, geometrical, mechanical properties, and any other related functionality factors. Architects, structural engineers and system engineers are all involved in such processes at construction design

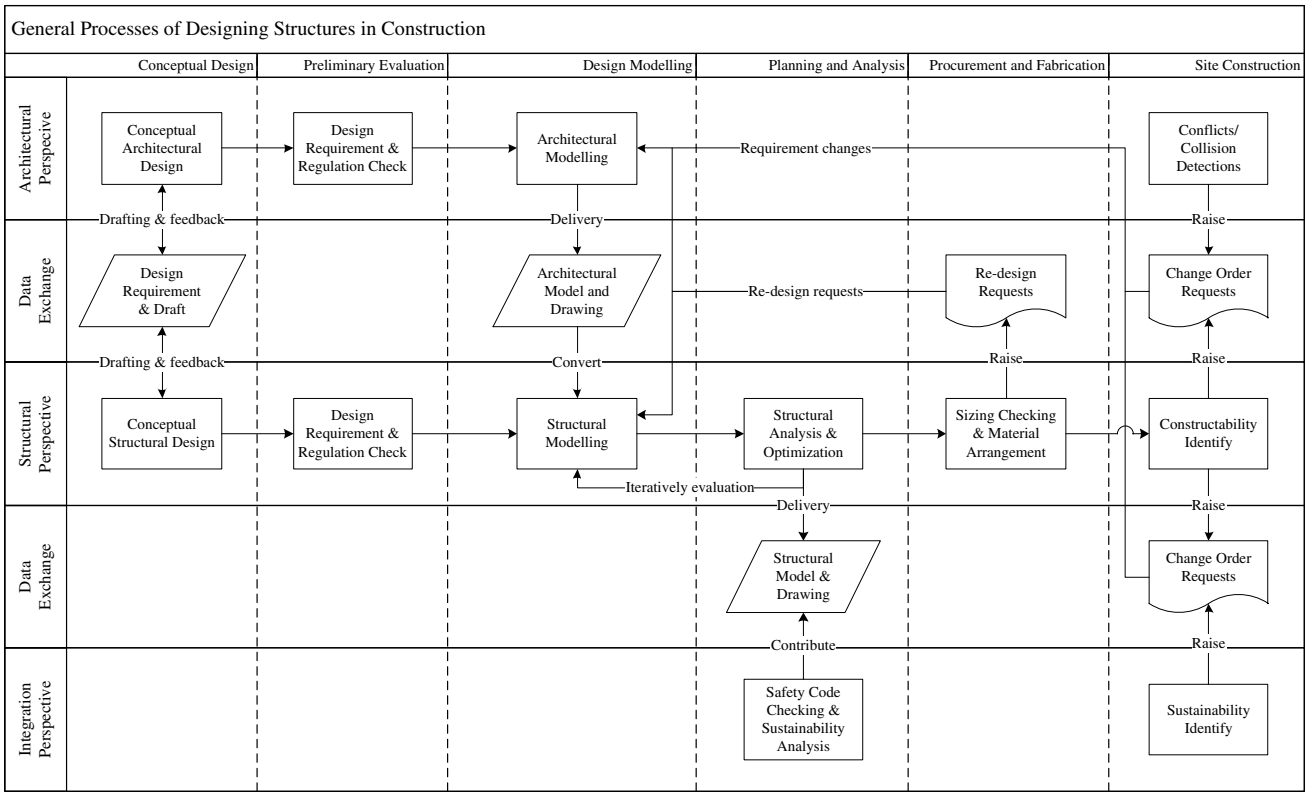

Fig. 1 Cross-functional flowchart for a general structural design in construction

phase. General processes of structural design can be seen as the following sub-sections, and particular state-of-the-art technical surveys are focused in three areas of the whole processes. They are also described in the later three sub-sections.

2.1 General Processes of Structural Deign

The general processes of current structural design can be seen in Fig. 1. In general, a structural case, such as a building design, starts with conceptual formulations at architectural and structural perspectives. Consideration to customers' design requirements, design theme and basic mechanical principles, architects can begin to draft design solutions for preliminary evaluations. The evaluations include self-checking for maintaining consistency of design ideas, feasibility tests and so on. Once the ideas are proved to be feasible, the detailed architectural modelling step will start. The outcome of architectural models specifies geometrical, and appearance properties, sometimes with a certain level of structural suggestions, they are delivered as architectural models, or drawings. The focus of the design task will then be moved to structural perspective.

By utilising the architectural models, or drawings for the following structural analysis design step, operators need to simplify models or drawings to only keep their mechanical meaning, such as beam-truss structure in steel frame cases. The converted models then can be analysed for their stability throughout structural analysis, and the results can be further improved by structure adjustments in optimisation processes. Usually skilled and experienced structural engineers make the selection of superior alternatives, or the decision of adjustments. According to state-of-the-art structural optimisation principles, they are iterative processes which terminate once the design objectives are met. Afterwards, structural-optimised models, or drawings can be proposed for furthering procurement and fabrication steps. Conventionally, evaluations of sustainability, safety, and other functionality of the designed structural models are also conducted at integration perspective. They usually involved in the late structural design steps and are treated as additional concerns for assisting structural optimisation.

Even after procurement and fabrication, the structural design plan can sometimes force changes due to manufactory, and material arrangement issues. Re-design requests can be raised for architectural and structural reconsiderations on current design, and suggest necessary modifications. During the actual construction phase, change order requests are common. Many reasons, such as conflict/collision detected, constructability and sustainability issues, can cause the demand

for change and shows that the flexibility of structural designs are essential during the whole life cycle of construction, and should be given particular attention.

Among these design processes, we put focus on three essential areas and identified their development status today. They include parameterised modelling processes, structural analysis and optimisation, and functionality evaluations at integration perspective. These will be described in the following sub-sections.

2.2 Parameterised Modelling Processes

Architectural and structural modelling processes are becoming configurable and flexible due to the efforts in parameterising the factors involved with design decisions, including geometrical properties of model components, design constraints, and so forth. The utilisation of the parameterised modelling process guarantees efficient responses for design changes, modularising design functions or instances, and ease of reuse in other similar projects. As a rapidly growing scale and complexity of current architectural and structural design projects, designers often suffer from serious organisational issues in handing design intentions during numerous design steps of modelling processes. Even with the advantages of computer-aided design, design conflicts and inconsistency issues among the increasing numbers of cost, scale, and construction elements need to be carefully addressed. Certain design management mechanisms [7] must be integrated to help organise design ideas during the modelling process in such a way that the result is either still consistent with the design ideas, or that modifications are based on an explicit modification of an earlier expressed idea.

Recently, parametric modelling [8–10] has made great progress in overcoming the above-mentioned issues. Unlike traditional modelling methods, parametric modelling puts more focus on formulating design entities in a parameterising way. In the parametric modelling process, designers need to not only produce their designs of geometrical objects, but also define variables that embed certain levels of design intentions. By manipulating the user-defined "dimension" of geometrical entities, designers can efficiently explore many possible design solutions [11], thus shortening the time required for a redesign [1] and helps increase the productivity of projects [11]. Furthermore, the formulation of a design concept becomes a serious topic, [10,12] due to the nature of modelling in a parametric modelling method. For now, parametric modelling has extended to not only representing parameterising processes by the meanings of its terms, but also using its parameters for different design meanings among geometrical objects [10].

The integration of parametric modelling in commercial markets is showing promising results. Generally, the first parametric CAD system is considered to be ProEngineer™ [13]. GenerativeComponents [14], a parametric modelling design system running on Microstation, provides the ability to capture design intentions among geometrical elements, and the formed graph structure, also called the generative component, can be easily reused for different scenarios by manipulating parameters in them. In addition, a Rhinoceros (Rhino) CAD system plug-in, named Grasshopper [15], has been developed with a well-defined graphical interface, supporting associative programming for parametric modelling processes. An ongoing commercial application is under development running on AutoCAD System, called DesignScript [16]. A script language with an elaborate syntax is available, and designers can build their models and, at the same time, keep readable records on design intentions inside the scripts. This is similar to including design ideas into a programming code. This completely changes the approach of modelling design.

With the benefits of parametric modelling methods, the design modelling results are improved. However, challenges and limitations still exist. The acquired design knowledge with higher levels of formulation becomes difficult for a system to interpret [10] and fails to provide necessary feedback. The act of parameterising models increases the complexity of a design [9,10], not only must designers make regular design work, they must also pay attention to the logical and internal structure of the design model. Sometimes, modelling tools based on conceptual aspects might completely change the modelling process. Designers need time to learn and adopt these tools instead of applying traditional modelling techniques. Furthermore, an efficient and precise way to form the structure of design work remains under investigation [9], and there should be common agreement to describe design intentions in order to enable communication between designers or collaborators [10].

2.3 Structural Analysis and Optimisation

Structural analysis indicates a serial of estimating processes for determining the stability and reliability of the designed plan provided in the structural aspect. Due to the invention of finite element method (FEM) and development of computer technologies since 1960s, the optimisation scheme has been introduced into engineering design for shortening the times of design cycles to achieve design objectives [2]. FEM has been widely utilised and successful in simulating system responses at many industrial fields such as aerospace, biomedical, chemical, manufacturing engineering and so on [17]. The analysis conducted by FEM, called finite element analysis (FEA), is generally divided into three steps: preprocessing, matrix calculation and post-processing [18].

Conventional structural analysis involves different professional roles, as shown in Fig. 2, and includes owners, designers, operators and analysts. Once the owners proposed their

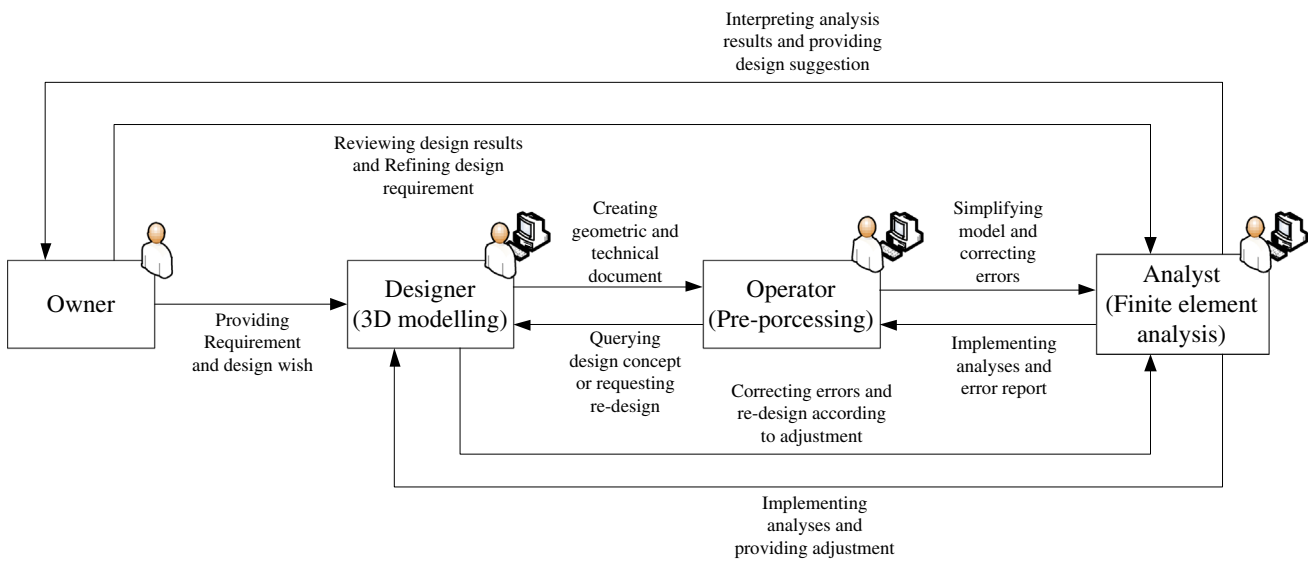

Fig. 2 Roles involvement in a typical structural analysis process (modified from [18])

design requirements to the design team, the designers, usually architects, should model the appearance of design solutions in terms of the art, geometrical and spatial design principles. Following this the designed models should go through a simplification process in order to preserve the structural meanings, such as loading component and joint connections, for the related structural analyses. These are pre-processing steps for the structural analysis. Operators should collect the models and technical documents from designers and sometimes consult with designers for getting design concepts and making error corrections. Once the model converting process has been finished by operators, analysts will perform matrix calculations and post-processing steps of the structural analysis. In the meantime, analyst should cooperate with designers and operators if there is any design issues found during the analysis processes. The results of the design solutions regarding, stability and reliability, will be shown, and the analysts will need to interpret them, and provide adjustment suggestions to designer, or to owners, in case the results do not fit into their requirements. These represent key works in determining the quality of a final design, and the satisfaction of the owner.

While structural analysis is taking place as computational methods for assessing design plans, certain search mechanisms or patterns are necessary for assisting structural engineers to find more appropriate solutions in order to guarantee the quality of the design. It thus becomes a structural optimisation problem. Typical structural optimisation problems include three categories: sizing optimisation, shape optimisation and topology optimisation [19]. Sizing and shape optimisation represent the processes of determining an area, and shape, of structural element with certain objectives, such as minimum fabrication cost, maximum structural reliability and so forth. Topology optimisation puts emphasis on global perspective for determining the connectivity of structural elements. The three optimisation problems were usually solved individually [20] until current research efforts showed that simultaneous optimisation results are better solutions [21].

The approaches of structural optimisation can be generally divided into two categories: gradient-based and stochastic methods [22]. Gradient-based methods require the evaluations of gradients of objective and constraint functions formulated from problems. Usually the structural problems are forced on dealing with nonlinear constraints on displacement, stress and critical buckling loading. Such approaches like nonlinear programming (NLP) [23], optimality criterion (OC) algorithm [24] and sequential quadratic programming (SQP) [25], have been developed and have achieved remarkable results. Stochastic methods do not need gradient information to converge to solutions instead of imitating physical or biological events by evolution of near-optimal solutions with iterations. The majority of these focus on heuristic approaches, such as genetic algorithm (GA) [26], evolutionary strategies (ES) [27], simulated annealing (SA) [28], ant colony optimisation [29], to name a few. They are widely used but usually consume considerable amount of computational power.

The current research for structural design optimisation problems of the construction sector can be seen in Table 1. In this research, we surveyed a limited scope and numbers of the state-of-the-art research works and concluded the current research interests accordingly. More detailed surveys can be retrieved by the work of Hare et al. and Habibi [19,22]. In general, recent research topics include structural reliability, bridge design, truss, steel frame and reinforced concrete structures optimisation. The majority of optimisa-

Table 1 A limited scope survey for conventional structural optimisation research

Research topic	Methods	Type	Description	Research
Structural reliability	GA	Stochastic	These research works focus on improving the efficiency of GA by integrating linkage-learning process and/or shredding operator to help reduce convergence time of optimisation processes for estimations of structural reliability	[89,90]
Bridge design structure	GA/ES	Stochastic	The optimised decision-making of bridge deck repair and maintenance is investigated by using GA. In addition, optimisations of bridge design concerning mixed design variables, shape, size, and topology of bridge are studied for minimising weight of bridges	[91,92]
Truss structure optimisation	GA/ES/PS/HS/SDS MIP/SLP/SQP	Stochastic Gradient-based	Optimising truss structures with minimum cost and weight has been studied and surveyed by enumerated studies, and will be continuously considered in the future. Trusses are essential structures in most of structural design cases for going through typical structural analysis, and are used as typical examples for comparing different optimisation methods proposed in these studies	[22,92–99]
Steel frame optimisation	GA/HS/SA/AC SLP/SQP	Stochastic Gradient-based	Similar with truss structures, steel frame optimisation is quite an important and widely-used task for optimisation studies nowadays. The research topics considering warping effect, seismic, section design, and weight minimisation of steel frames have been studied. Researchers usually implement and compare these optimisation approaches by using steel frame cases as well	[22,98–104]
Reinforced concrete frame	GA/PS	Stochastic	Comparing with steel frame structures, reinforced concrete structures can be more complicate because they are combinative materials, and their costs are not simply estimated by the weight of them. In addition, they can be designed with various patterns of reinforcements. Current researchers target the topics to minimise total cost of reinforced concrete flat slab or planar structures	[105,106]

GA genetic algorithm, *ES* evolutionary strategies, *PS* particle swarm, *HS* harmony search, *SDS* simplicial direct search, *SA* simulated annealing, *AC* ant colony, *MIP* mixed integer programming, *SLP* sequential linear programming, *SQP* sequential quadratic programming

tion approaches are developed based on heuristic methods in recent years, which shows the increasing interests of stochastic approaches in solving current structural optimisation problems. This may be due to the discrete and combinatorial natures of structural design problems and a global optimum is required. The objective functions defined in these research works usually target on minimizing cost and structure weight, and simultaneous optimisation have been concerned within them. According to the trend of the surveyed structural optimisation development, future developments can focus on identifying the efficiency of different optimisation approaches, considering hybrid structures and inventing or modifying existed heuristic approaches for better usages in particular structural design domain.

In the state-of-the-art structural design, structural optimisation processes are usually rarely applied, because it is time-consuming and inflexible to deal with numerous variables in an actual structural case [30]. In addition, the structural analysis itself can only generate simulated results at mechanical or sustainability aspects, such as the status of displacement, stress or defects of structures, which usually does not contain information for selecting superior design alternatives, or lacks leading supports [30]. In particular, the strategy and methodology of selecting design alternatives is one of the key steps during optimisation and needs to be developed further to efficiently apply to structural design works [31]. Conventionally skilled structural engineers have to lead the directions of interpreting analysis results and make design adjustments upon their experiences around iterative design processes. Under such circumstances, the flexibility and design ideas of structural design works will be limited after the architectural and structural models are established [32] and going through time-consuming and tedious analysis steps. Due to the extra works and potential information

loss on simplifying models for different purposes [18], these inflexible situations derive even worst results and reworks during change order stage. Thus, the structural design analysis, tools and considerations, such as functionality and safety, should be adopted earlier during the conceptual design stage to make sure optimised solutions can be discovered and to maintain the abilities to redesign.

2.4 Functionality Evaluations at Integration Perspective

Apart from analyses with mechanical purposes, functionality issues of structures concerning structural regulations (other than structural stability), safety, sustainability and environmental factors are usually addressed during structural design. Those issues can also influence the topological optimisation of structures and conventionally be evaluated in the late structural analysis and optimisation stage.

The rules regarding safety and structural regulations were usually generated and interpreted by human input. Their structures were often complex and sometimes incomplete because of the difficulty of exhaustive enumerations. Therefore, improving the logic of regulatory codes has been researched [33]. In the meantime, some research was undertaken to combine rule-checking with structure representations for fire escapes and evacuation purposes [34]. Afterwards, simulation approach of rule-checking for human activities, such as accessibility, has also been surveyed and developed [35]. All the rule-checking innovations are still in a developing stage, which are needed for further investigations.

As for sustainability evaluations, the environmental cautions are getting significant with increasing demand of sustainable concerns. The agency for environment and energy saving (ADEME) reported that every year the building consumes 43 % of total energy consumption and emits 24 % CO_2 of national emissions [36]. A major consumer of energy is residential structures, which represents 30 % of total energy consumption in average worldwide [37]. Current research has shown their contributions with respect to achieving low-carbon and environmental economic purposes. Simulations of energy requirements of buildings and building energy analysis (BEA) [38] have been undertaken during design stages.

Current issue of functionality evaluations is the late adoption during the whole structural design process. Sometimes they are performed after the design stage [39]. The solution in the structural analysis and optimisation step is limited to existed architectural or structural design models, which specifies the general design intentions. The adjustments according to functionality evaluations are sometimes compromised with design successors' restrictions from structural aspects, or otherwise, makes tedious and time-consuming reworks for the entire design process. In terms of current development, the greatest opportunities for implementing sustainability and environmental considerations are at the conceptual design stage [40]. In addition, there is a significant amount of variables needed to be concerned with during the integration level of structural design, such as sustainability, environmental factors, or other functionality elements necessary to consider. How to organise and manage such massive data and keep the design processes as simple as possible is an essential topic for further investigations. Cooperation among design teams with different professions, highly quality communication and uniformed data exchange mechanisms are all demanded among such design processes.

3 BIM Integrations in Designing Structure

With the implementation trends of building information modelling, model, and management technologies, the majority of current construction projects have adopted these ideas in the entire life cycle of construction, including the design phase. The following sub-section describes the features and benefits which bring BIM as an innovative platform in designing structures according to the surveyed evidences. In particular, four state-of-the-art development directions related to structural design: parameterised building database, visualization and interactive modelling, communication and data exchange, and sustainability evaluations and code checking, are all surveyed in the next four sub-sections.

3.1 The Benefits of BIM in Designing Structure

Utilising building information modelling technologies for reshaping current structural design has been widely-adopted in construction practice. With systematic modelling processes, powerful interactive visualization platform, and standardised data exchange interfaces, BIM-enabled structural design has improved its performance and reduced cost during the whole life cycle of construction [30]. Consider the general process of structural design shown in Fig. 1. BIM can be a design information centre for handling massive data and different roles involved in the structural design processes, which helps provide necessary data at the right moment, and speed up communication and cooperation time through its visualization and interactive abilities. In addition, functional concerned variables of design projects can be collected in uniformed database and various evaluations regarding safety or sustainability can be performed with pre-defined relationships among designed structures. It helps personnel decrease the possibilities of making design disorders or errors in between different design teams.

Considering the benefits coming from adopting BIM, over 82 % of BIM users believe that BIM has positive impacts on their company's productivity [5]. Stanford University's Cen-

tre of Integrated Facilities Engineering (CIFE) reported that using BIM in construction can provide up to 40 % elimination on unbudgeted change, controlling cost estimation accuracy within 3 %, and reducing 7 % of overall project time [5]. The case studies demonstrated in Azhar's research [5] show that the return on investment (ROI) of BIM can be significantly higher than traditional construction projects, which depicts potential economic benefits. Nawari [41] also shows promising successes of BIM on creating more sustainable communities. According to McGraw-Hill Construction's report [42], architects gain majority benefits by implementing BIM among construction life cycle. As for organisation aspects, the learning processes for employees on BIM design tools are necessary for design and consulting companies. And the initial cost and labour hours for implementing BIM should be devoted [1,5]. The Unite Kingdom government requires the adoption of collaborative BIM in all public projects by 2016 [43]. The benefits of BIM in improving structural design practice include but are not limited to: reduction of design and drafting errors, increasing design productivity, flexible design and reduction of change order costs, and unified data exchange formats.

About implementation status of current BIM in designing structures, we have established a survey focusing on many research directions among the design processes. The general descriptions can be seen in Table 2. Current BIM-enabled structural design applications can be categorised, but not be limited, as the following topics: parameterised building database, visualization and interactive modelling, uniform communication interface, and sustainability evaluation and code checking. The current issues of structural design concerning above-mentioned categories have been identified and how these issues are solved or improved by BIM technologies is listed. They will be described individually in the following sub-sections.

3.2 Parameterised Building Database

The idea of building information model is to capture any beneficial properties with the model components themselves for utilising, determining, and evaluating at the whole stages of construction projects. These demands can be achieved by parametric modelling and data integration processes. Eastman et al. [3,44] reviewed the transition from 2D drafting to 3D modelling and BIM development, which includes parametric modelling among complex geometrical structures. Sacks et al. [11] developed a language for describing geometrical dependencies in a parameterised way. It is expected that the concerned properties should include not only geometrical information to help constructability but also any functional variables (e.g. safety level, structural robustness, redundancy, ductility [30]), environmental (e.g. energy saving for heating, ventilating, and air-conditioning (HVAC) [45]) or social interactive data (e.g. serviceability, evacuation instructions [46]). And that is the entry point for adopting parametric modelling technologies to quantify those variables into a building information model, and make their maximum usage by stakeholders or computers for increasing productivity, efficiency, and decreasing cost during design stages.

The related research of parametric modelling and integrative BIM database starts on various and diverge directions, including assisting rebar optimisation processes [47], reinforcement placement [48], formulating building object behaviours as controllable parameters [10], and so forth. Aram et al. [49] developed information flow processes model for concrete reinforcement's supply chain, which identified how BIM tools can support and improve the conventional supply chain. Diao et al. [50] proposed an optimal design-aided system based on the concept of BIM that put database, environmental simulation, and optimisation algorithm together. Hu et al. [51] proposed a structural analysis method which can automatically generate structural models with geometric, resistance, and loading conditions, by links of the schedule, architectural models, and material properties. Cerovsek [52] proposed a complex adaptive system with BIM (CASB), which contains knowledge database and can be used to automate the transition between the architectural design and construction technologies. All these research works show that the efforts to maximum the use of BIM as central data processing units are continuously engaged, so does the development of BIM interoperability.

3.3 Visualization and Interactive Modelling

The BIM technologies can help designers visualize their structural design results and make real-time adjustments in an interactive environment. It is all because one of the design intentions for BIM that is to provide sufficient and intuitive information to related personnel during construction life cycle. And a significant amount of BIM visualization tools (e.g. Autodesk Revit [53], Tekla Structure [54]) target these features for users to interpret design model more easily. Immersive equipment, such as head-mounted display [55,56], gesture control [57] gloves, and visualization methods, such as virtual reality (VR) [58], augmented reality (AR) [59–62], mixed reality (MR) [55], and so on, also enhance the understanding of designers for identifying their design errors or points of interest. Current finite element analysis (FEA) applications have been developed with integration of BIM visualization tools, which provides interactive abilities for frequently design adjustments, and ease of finding design errors among the analysed geometry. Take Autodesk Revit as an example, Table 3 shows a preliminary survey for FEA applications, which provides rich compatibility and can be executed simultaneously with Revit. The development trends of such integrations show that BIM visualization abili-

Table 2 Enabled technologies and research for current BIM-enabled structural design

Direction	Conventional design issues	Approach	Research
Parameterised modelling	Inflexible design results in a significant amount of the re-design cost	Parametric modelling (e.g. functional, environmental or social interactive data)	[10,47–52]
	Integrations lack concerning functional design variables	Integrated database system	
Visualization and interaction	Design errors and disorders are not easy to be found	3D visualization tools with navigation	[30,63,64]
	Analysis tools of structural design lack rapid adjustment abilities	Immersive equipment	
		Virtual reality (VR) / Augmented reality (AR)	
Communication interface	Inefficient communications exist different roles during structural analysis processes	Integrative and interactive platform	[65–68]
	The use of current design tools lack intuitive and cooperative abilities	Internet and cloud-computing technologies	
Data exchange	Information lost among the convert processes of models with different purposes	Standardised data exchange format (e.g. IFC, LEED, DGNB)	[43,69–73]
	Information fragments and disorders exist among different design teams	Uniform interface design	
Sustainability evaluation	Evaluation after modelling processes limits the flexibility of the design plan	Integrated database system in structural design processes	[74,75]
	There exists an increasing demand of considering sustainability on current structural design	Sustainability evaluation packages with BIM software	
Code checking	Labour-intensive and exhaustive checking works are needed among different regulations and codes	Automatic code checking	[78–80]
	Decision-making processes of the design are not efficient and they are without sufficient instructions	Rule-based BIM database	

ties indeed necessary for improving current structural design processes, and they are worthy to further investigations.

As the benefit comes from intuitive visualization platform, collision, or conflicts among models at different design perspectives can be found in a semi-automatic or an automatic way, and to further support the evaluation of designed structures. Integrative structural design system is developed by Gerold et al. [30], which targets on facilitating interactive modelling processes and real time adjustment for structural analyses. Yan et al. [63] utilised the concept of BIM and games to create an interactive design-play environment for the architectural design and education. Wang et al. [64] developed a conceptual framework to identify the augmented reality integration with BIM for assisting designers to on-site communications. These research works can raise essential development focuses for future researchers' reference.

3.4 Communication Interface and Data Exchange

By taking advantages of BIM, as an interaction media, it contains more communicative opportunities for personnel in between different structural professions, and it helps establish stronger cooperation among design teams. With integrated interactive abilities provided by BIM, the opportunities of cooperation with different design professions are increased, even with clients [65]. The information on BIM database can be shared and discussed through Internet, and with cloud computing technologies to ease of accessing from corresponding design team members. Redmond et al. [66] have conducted interviews of 11 experts about using cloud-BIM applications, and found out that the investigated applications would enhance BIM usability experiences in making design decisions. Singh et al. [67] identified technical requirements on using BIM database as a multi-disciplinary collaboration platform, which points out the configuration and implementation support are critical for adoption and enhancement of such technology. Shen et al. [68] applied building information modelling, user activity simulation, and requirement management to develop an evaluation method for enhancing designer-client communication, especially at early conceptual design stage.

Data exchange abilities are also one of BIM's objectives for tackling information fragmentations among design teams. A typical standard is the Industry Foundation Classes (IFC)

Table 3 Current FEM analysis software with the integration of BIM tools (taking Revit Structure as an example)

Name	Organisation	Type	Description	Reference
Advance Design	Graitec	Application	The application is developed for the structural analysis of the reinforced concrete and steel structures. It includes a FEM engine and performs structural analysis according to Eurocodes, North American, and Canadian codes. It contains compatibility features that the data can be exchanged through IFC, CIS/2, SDNF, and PSS interfaces with Autodesk Revit Structure	[107]
PKPM	China Academy of Building Research	Application Package	The application package is widely-used in China as required tools for architectural and structural design projects. The tools in this package, such as SATWE and FEQ, are capable to be used for high-rise building, shear walls structures analyses. These tools also provide Autodesk Revit plug-in for converting models	[108]
FEM-Design	StruSoft	Application	The application is developed based on Eurocodes, and is used for analysing on concrete, steel, and timber structures. It provides add-ins for Revit Structure, ArchiCAD, and Tekla Structure so that the analysis model geometry generated from these applications can be transferred into this application for further analyses	[109]
2D/3D FEM	SOFiSTiK	Application Package	These applications provide finite element analyses that support various kinds of structures, including 2D, or 3D slab, shell, beam, shear walls, and so forth. They can be used for building, bridge, and industrial structure design. The data formats are all compatible with that of AutoCAD and Revit Structure	[110]
Robot Structural Analysis	Autodesk	Application	The application is an Autodesk product, which combines with Revit Structure	[111]

[69,70], which provides extensive domain-specific information structure for BIM. With current design projects moving towards to BIM adoption, it is essential to various architectural, structural, or functional analysis tools to be interpretable with open BIM schemes like IFC [43]. As for sustainability aspects, leadership in energy and environmental design (LEED) standard issued by the U.S. green building council [71], the handbook published by the German sustainable building council (DGNB) [72], and file formats like the green building XML (gbXML) [73] standard for the performance-oriented design, are all proposed tending to unify and establish the assessment principles for the current structural design. Numbers of commercial finite element analysis applications have been developed according to those standards, and these applications have further built stronger connections with BIM visualization tools. For example, they are like the ones shown in Table 3.

3.5 Sustainability Evaluation and Code Checking

There has been a significant amount of research studied on the sustainability of structures in recently ten years, which accounts on BIM-related information management. According to the survey of Becerik-Gerber and Kensek [4], 83 % of practitioners (110 in total) are agree that BIM for the sustainable design/construction is potential area for further investigations. By utilising a uniform BIM database, which integrates various design variables regarding the sustainability of structures, corresponding designers can easily retrieve the necessary data and eliminate design conflicts against sustainability concerns. To name a few, König et al. [74] prototyped an open knowledge-based system, and made it possible to capture, analyse, and share information of building sustainability among the stakeholders. Wu and Chang [75] developed a ratio of equivalent transparency (Req) calculation tools considering object-oriented modelling, and parametric design. In addition, numbers of functionality evaluations, such as computational fluid dynamic (CFD) [76], energy saving [77], and so on, have been encapsulated as packages or compatible applications to integrate with current BIM visualization tools. They can analyse and provide rapid analysis results through certain model simplification processes. As a preliminary survey that can be seen in Table 4, there are an increasing number of CFD applications which integrate with BIM tools. They can exchange analysis results by standardised data format, such as gbXML, and so forth.

Code checking helps designers by through appropriate tools and knowledge based on information stored in building information model, so that designers can not only concern design quality at mechanical aspects, but also at safety,

Table 4 Current CFD analysis software with the integration of BIM tools

Name	Organisation	Type	Description	Reference
Simulation CFD	Autodesk	Application	The application is an Autodesk product, which combines with Revit Structure	[111]
blueCFD-Core/AIR	blueCAPE	Application Package	The application provides CFD analyses for the AEC community. The application is established based on an open source CFD package called OpenFOAM. The input data format is gbXML, which is compatible with other BIM applications or other LEED applications for structural and energy information exchanges	[112]

sustainability, and functionality aspects. The development of industry foundation classes (IFC) [70] in the 1990s has led research to use building model schemes for code checking. Automatic code checking has been realised in recent years for assessing the building designs according to various criteria of design concerns. According to Eastman et al.'s definition [78], the automatic rule-based checking generally includes the following processes: rule interpretation, building model preparation, rule execution, and rule check report. Zhang et al. [79] developed automated safety rule checking algorithms for BIM, which detects safety hazards by going through rule interpretation, mapping to building information model, and rule execution steps, for generating graphical reports. Martins and Monteiro [80] developed an application for performing automatic code checking in domestic water system. It is believed that in the near future the benefits of such approaches adopting in the current structural design practice can be revealed.

4 Impacts of BIM-Enabled Structural Design

With the increasing number of successful applications, BIM-enabled structural design can be continuously implemented and improved in the near future of construction practice. The impacts of introducing BIM considerations, concepts, and modelling/evaluating mechanisms in the current structural design, include but not limit to: the awareness of systematic concerns in designing structures, integration/cooperation demands for design teams, potential fragments among communication and data exchange interfaces, and interactive abilities during design processes.

4.1 The Awareness of Systematic Concerns in Designing Structures

The aim of building information modelling is not only considering static variables, such as geometrical restrictions, or material properties, for structural or architectural design. It is also considering dynamic and interactive factors that influence the overall design performance during construction life cycle. The designers involved in structural design should be aware of the usability of the building, such as functionality, safety, and sustainability issues, other than the conventional structural criteria. Designers are forcing to consider the whole pictures of the construction plan, rather than their own profession areas. The development of general formulation mechanisms for design variables is necessary, and should start up to be investigated in order to quantify dynamic and interactive factors, and helps designers make better decisions for dealing with multi-objective design problems.

4.2 Integration/Cooperation Demands for Design Teams

Due to massive considerations of deign variables comparing to traditional structural design methods, different design teams in a BIM-enabled structural design project with different professions, such as architects, structural engineers, and MEP system planners, need to have strong connection with each other to cooperatively solve critical issues, and avoid design conflicts. It is getting essential nowadays especially in large-scale construction projects. For achieving better design solutions than that of the conventional structural design, the trade-off can be the increasing complexity of the design project, if the related assistance tools are not ready, or still need to be investigated. The level of the cooperation among teams can also influence the successful rate a design task could achieve. The consequential relationship is enhanced among the BIM-enabled structural design practice.

4.3 Potential Fragments Among Communication and Data Exchange Interfaces

With the massive cooperation among design teams, fragments of communications and information loss during data exchange processes become essential issues in the BIM-enabled structural design. Design teams and related BIM tools development face such problems by a highly chance than usual. Suitable interactive mechanisms or novel visualization technologies during discussion, such as augmented and virtual reality [81,82], dynamic interactive modelling [30], expert system for interpreting structural analyses

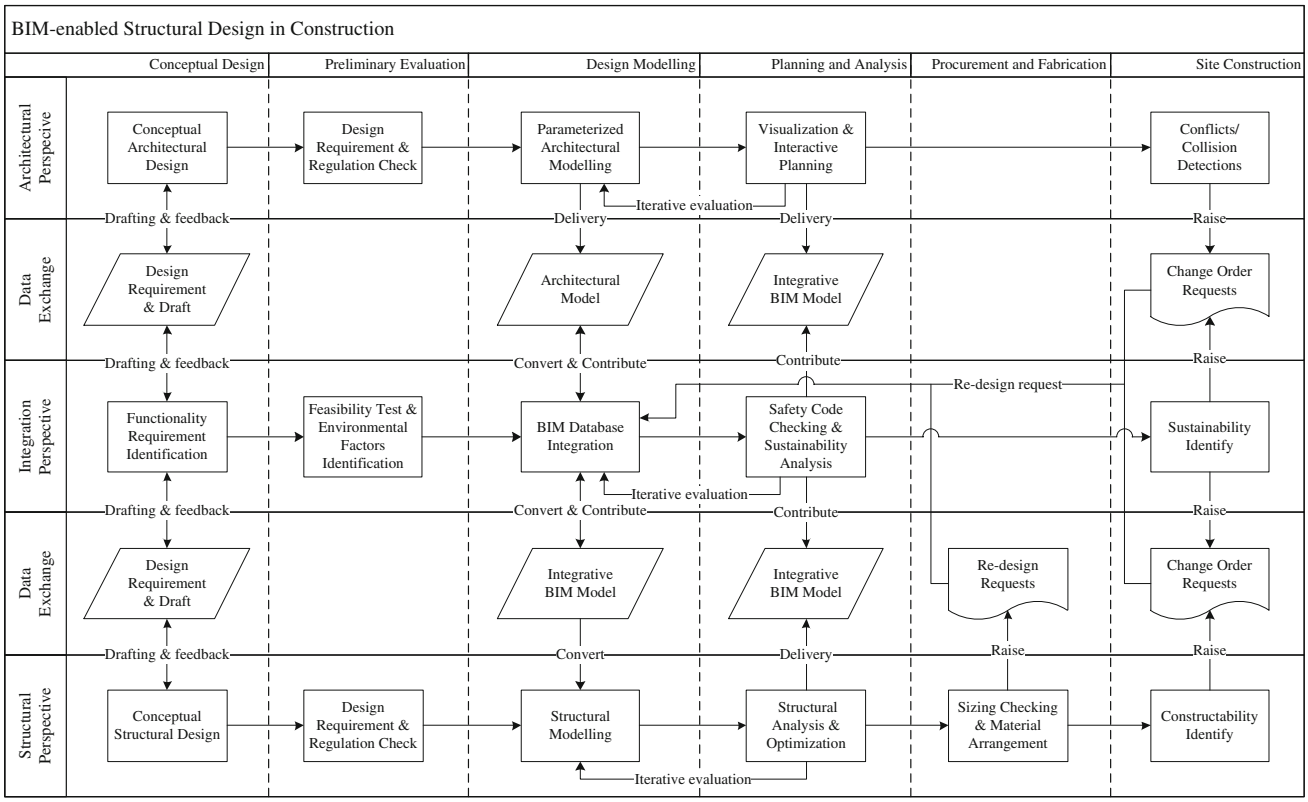

Fig. 3 The expected cross-functional flowchart for a BIM-enabled structural design

results, and so on, are getting to replace the sole structural design methods. Standardisation demands of data format [52,83] become important research topics for eliminating such interfacing gaps.

4.4 Interactive Abilities During Design Processes

By implementing BIM concepts in designing structures, real time interactions with structural models during analysis and evaluation steps become possible taking advantages of BIM visualization tools. For maximising benefits of time-efficient design processes, or communications, designers are obliged to discover new way increasing interactive abilities among team members, or design models. Utilising novel visualization platform, such as BIM 3D modelling tools [53,54] with various properties on structural components, designers have more flexibility to make adjustments on structures, and to visualize the results through analyses.

5 Future Trends of BIM-Enabled Structural Engineering

As BIM-enabled engineering is realising nowadays, future developments and applications which benefit to the structural engineering are expected with the supports, such as increasing but economic computational power, parameterised drawing software, real time and immersive 3D visualization equipment, and so on. We summarise the surveys by pointing out the development directions in the future BIM-enable structural design processes and potential applications. The expected general process in the near future can be seen as Fig. 3.

In the refined design flow, design processes at integrative perspective become dominative thread which provides unified and parameterised BIM database for the use of architectural, structural, and other functionality concerns. All of design concerns should be considered, and cooperative efforts should be devoted at early stages, such as at conceptual and preliminary evaluation stages. Design personnel with different domain knowledge can take responsibilities for preliminary evaluations at different perspectives, such as design requirement and regulations checking, feasibility test, environmental factors identification, and so on. In parallel, all initial works should parameterised necessary considerations so that the searching space of design solutions can be expanded, and maintain its flexibility during the modelling, or the rest of the processes.

During the design modelling phase, novel parametric modelling technologies [16] for generating architectural model are expected to be implemented in order to integrate geometrical, functionality, and structural parameters into a uniform BIM database. A BIM database is supposed to pro-

vide general interfaces for different level of detail (LOD), and different analysis purposes, to easily coverts necessary information, and efficiently gets feedbacks without errors and data lost. Thus, all planning and analyses at architectural, structural, and functional perspective can be performed iteratively based on the same information delivery and exchange platform. The planning and analyses can include but not limit to visualization, interactive planning, code checking, sustainability analysis, and structural analysis and optimisation. The cooperative design works will help decrease the chance of design errors, disorders, and logical conflict issues, and at the same time increase the quality of optimisation results. Once an optimised solution is decided, and the construction project moves into procurement, fabrication, and site construction phases, any re-design request should be fed back to integrative BIM database, and let corresponding design teams to handle. Every consequential change will be notified due to the well-established dependency relationships among geometrical components, and functionality subjects of the BIM database.

In particular, five future trends of the BIM-enabled structural engineering are addressed as follows:

Parameterised structural design regarding structural functionality, sustainability and safety : According to the trend of the parametric modelling development, conventional geometrical parameters for the architectural modelling could extend their definitions to include other functionality, sustainability, and safety-related factors, and these can be potentially widely-utilised in structural design. According to Sacks and Barak's work [1], the potential productivity gain of adopting parametric modelling is approximately in the range from 15 to 41 %. By considering above-mentioned factors as standard design processes, the flexibility of designed structures can be assured in handling different construction requirements, not just for the structural satiability and the cost. The building information model conducted by such processes guarantees change orders at actual construction stages, or design evaluation meetings could be efficiently executed without costly, tedious, and time-consuming works. It can be a trend for exploring more parametrical approaches to quantify more properties of the building information model for the ease of generating design alternatives and redesign processes. As for evidences supporting such predictions, research efforts on reducing the difficulties of parameterising processes and standardising the procedures with regard to domain-specific design cases can be seen in Lee et al.'s [10] research. Related commercial software [16] has being developing, and innovation modelling methods which focus on capturing behind design intentions rather than geometrical properties have also been emphasized. It is expected that modelling processes and the training ways of designers will have revolutionary changes in the near future.

Early adoption of structural optimisation at conceptual and interactive modelling stages : The design variables of the structural optimisation, such as safety, functionality, and sustainability will be considered at early stage of the conceptual design so that it maximises the design flexibility and helps explore larger optimisation solutions space with respect to different design perspectives. It works better rather than mainly, or only replying on experienced structural engineers. With sophisticated visualization details taking advantages of BIM, and the exponentially increasing amount of computational power, time-consuming structural analysis and optimisation processes would be performed during acceptable time periods. Distributed, or parallel computing mechanisms applied for structural analysis methods are promising approaches as well [84]. Appropriate design tools combined with BIM visualization platform could be developed for generating integrative structural design environment [30]. Novel modelling approaches, such as mock-up using 3D printer, can also be widely-utilised to support the better spatial cognition of designers in the near future. It can help young, and inexperienced structural engineers establish design senses, and sharpen the learning curve by trail-and-error processes through rapid, and efficient analysis feedbacks, and sophisticated visualization interactive environment.

The development of effective decision-making support tools based on BIM database and related visualization technologies : With regard to cutting edge technologies like current structural analysis and optimisation methods, they are rarely utilised in the large-scale construction practice, because not only their time-consuming natures but also lacks decision-making guidelines for conducting "better" solutions during iterative design processes. Engineers without enough design experiences may fail, or struggle to improve current structural design plans. The development of BIM-related tools which focuses on accessing parameterised functionality variables, and fast react upon design adjustments, and some rule-based BIM checking systems [78] can eventually provide meaningful hints for further design decisions. Automatic assessment mechanisms like Zhang et al.'s [79] research, which detects safety hazards by going through automatic rule checking algorithms, is a promising research direction as well. These are expected to support designers to their decision-making. Moreover, visualization provided by BIM tools can speed up error discover, and analysis interpretation processes for establishing rating schemes on design alternatives. It is expected to widely adopt visualization functions, or equipment during structural design processes.

High performance numerical methods for large-scale optimisation problems : It is certain without doubting that the development of numerical methods for structural optimisation will continuously plays significant role in improving

structural design quality. Under such prerequisites, conventional gradient-based and stochastic optimisation algorithms will encounter serious obstacles due to the corresponding solution time, and storage requirement are huge [2]. The high performance methods developed to deal with the modern construction practices are highly desired, even with the exponentially increasing of the computational power, and the development of the sophisticated interactive visualization nowadays. A fast response, and sensitivity re-analysis mechanism is promising research goal that the related research has already been devoted [85]. Developing efficient numerical methods, such as equivalent static loads for the nonlinear dynamic response structural optimisation [86], are also undertaken. GPU-accelerated FEM structural analyses also contribute to the efficiency of analysis processes [87]. Furthermore, multi-scale design and optimisation problems are still at developing stages [2]. And there are many research opportunities on solving dynamic, or non-linear structural problems with uncertainty natures for optimisation.

Strengthened cooperative works with data exchange abilities of BIM : Data exchange issues for BIM technologies still demand further investigations, and need to build standardised protocols around governments, owners, contractors, and related stakeholders at world-wide aspect [83]. Some researchers already put their focus on identifying the requirements for the information sharing among the AEC stakeholders [74]. The development energies and works will force fragmented interfaces, communication obstacles, and interpretation errors during structural design to be decreased, or eliminated, even such issues around the whole construction life cycle. The research focuses should be put on identifying the criteria for developing associated databases with unified interfaces, and eliminating the need of model covert efforts and reworks on different design purposes [4]. The early corporations among architectural, structural, or related system design teams can avoid unnecessary man-made errors, and to deal with the design problem in a systematic way. Most of construction-related organisations in UK agree with that collaboration is the most important key to successful BIM [6]. The information technologies (IT) supported collaborative works will continuous to be an area under intensive investigations [88]. With unified interfaces, and convert protocols in the BIM-enabled structural design, the quality of structural solutions will be increased, and significantly reduced redesign demands during design, or construction stages.

6 Summary and Conclusion

The modern management paradigm at current architecture, engineering and construction (AEC) field is moving towards building information modelling. According to such development trend, structural design processes inevitably demand a certain level of revolutions. And personnel involving in the whole design process should have an open mind to embrace innovative modelling methods, information management, tools, and applications coming with the adoptions of BIM.

In this research, the impacts and potential development trends of BIM-enabled structural design have been identified through a comprehensive survey regarding the structural design practice and current status of BIM adoptions. The general flow of current structural design processes has been established, including related processes at architectural, structural, and functional perspective. Without following BIM's design concept, conventional structural design lacks design flexibility in handling the re-design requests, and interactive and cooperative abilities at early conceptual design stages. The tedious and time-consuming structural optimisation processes still need serious research efforts to improve their efficiency by inventions of high performance analysis methods or integrations with increasing power of computers.

By utilising BIM to facilitate structural design practices, design thread with integrative concerns should dominate the whole process. It provides unified and parameterised BIM database for the use of architectural, structural and other functionality purposes. All design works at conceptual stage should parameterise necessary considerations so that the searching space of design solutions can be expanded and maintained its flexibility during the modelling or the rest of the processes. With increasing computational power and interactive abilities of BIM tools, the collaboration opportunities among different design teams will increase, and the optimisation processes will become relatively efficient works rather than tedious or time-consuming ones as before. The cooperative design works will help decrease the chance of design errors, disorder, and logical conflict issues, and at the same time increase the quality of optimisation results. A significant amount of research efforts is devoting to improve such ideas in designing structures including parametric and interactive modelling, visualization, uniformed communication interface and data exchange mechanisms, sustainability evaluations, automatic code checking, and so forth.

Impacts of BIM-enabled structural design can be concluded as: (1) The awareness of systematic concerns in designing structures; (2) Integration/cooperation demands for design teams; (3) Potential fragments among communication and data exchange interfaces, and (4) Interactive abilities during design processes. They are essential concerns which need to be considered by the personnel of structural design teams. The future development trends of BIM-enabled structural design are expected to be devoted at (1) parameterised structural design regarding structural functionality, sustainability, and safety; (2) early adoption of structural optimi-

sation at conceptual and interactive modelling stages; (3) innovative development of effective decision-making support tools based on BIM databases and related visualization technologies; (4) high performance numerical methods for large-scale optimisation problems, and (5) strengthened cooperative works with data exchange abilities of BIM. The research focusing on those trends is expected to be engaged in the near future.

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